# Saarthi: A Smriti – A Knowledge-Driven AI Robot for Intelligent Communication and Assistance

Rambabu Pemula[1], Anusha Ampavathi[2] , A. Obulesu[3], Suresh Babu Changalasetty[4] ,S. Ashwini[5] , O. Madhavi[6], M. Lakshman [7], N. Anil Kumar [8].

*Vidya Jyothi Institute of Technology, Aziznagar Gate, Chilkur Balaji Road, Himayat Sagar Rd, Hyderabad, Telangana 500075*

[1)] rambabupai@vjit.ac.in, [2)]anuampavathi@gmail.com, [3)]aihod@vjit.ac.in, [4)]sbabu@kku.edu.sa, [5)]ashwinisamaneti@gmail.com, [6)] odurimadhavi3366@gmail.com, [7)] m.lakshman98@gmail.com, [8)] anilbabulu711@gmail.com.

**Abstract—An AI-Enabled Multimodal Interactive Robotic System is introduced to support intelligent human-machine communication and real-time environmental monitoring. It integrates cloud-based large language models with embedded hardware to enable natural language dialogue, visual feedback, physical gesture and movement, and continual safety sensing. Using a Raspberry Pi 4 Model B as the main controller, it runs a multi-threaded Python framework to feature speech recognition, text-to-speech synthesis, web-based control procedure and sensor monitoring. It features an efficient motor controller, TB6612FNG which makes it easy to control the motors, and a SSD1306 OLED display to show animated facial expressions that confirm with the speech output. It features an active-high infrared flame sensor as a safety feature to detect unsafe conditions and trigger alerts. The experimental results demonstrate low-latency interaction and reliable communication to achieve a responsive emergency first aid. It is proof of concept that low-cost, intelligent service robots can be built for education, medical environments and the smart environment.**

*Keywords: Human-Robot Interaction, Multimodal Robotics, Large Language Models, Raspberry Pi, Service Robot, Environmental Monitoring*

## 1. INTRODUCTION

The development of sophisticated artificial intelligence (AI) and embedded systems have brought a revolutionary transformation of traditional robotic platforms into interactive machines powered by artificial intelligence. Traditional robots can only know how to act or move with the set commands they were programmed with, and none of them have ever been able to converse either with the users (humans) or within human language. But in recent years, large language models and cloud computing systems have also reached the point where robots can understand natural language, generate meaningful responses to user questions, and engage in social interactions with users.

Human-Robot Interaction (HRI) concerns the design of robots that can successfully communicate with humans. In addition to moving and performing tasks, modern service robots are expected to understand spoken words, generate responses to users, and ensure that users are safe. However, many low-cost robotic platforms do not have the capability to develop effective conversational intelligence and monitoring of their surroundings.

This paper describes the design of an AI-enabled, multimodal interactive robotic system that combines conversational intelligence, visual feedback, remote control (over the web), and real-time environmental monitoring of the robot's surroundings. The AI-enabled robotic system employs cloud-based artificial intelligence models for understanding and generating natural language, while embedded hardware is responsible for managing

the sensors and actuators associated with the robot's physical movements. Our goal is to create a scalable and affordable robotic platform that is suitable for use in indoor service applications.

## 2. LITERATURE SURVEY

Natural interaction with people has become a significant feature of service robots thanks to the rapid progress in robotics, artificial intelligence (AI), and multimodal interaction technology. Researchers have examined speech interaction, autonomous navigation, emotion recognition and intelligent conversational systems as ways to enhance human–robot cooperation. Different relevant research articles regarding the proposed intelligent multimodal interactive robotic model are reviewed in this section.

Spezialetti et al. presented an emotion recognition framework for Human–Robot Interaction (HRI) that comprehends human emotional states based on speech signals, facial expressions, and physiological information in the first reference [1]. The degree of emotional intelligence embodied by social robots has been emphasized in the study to improve their behavior. Integrated real-time communication and multimodal environmental monitoring are lacking in the literature; rather, this work focuses on emotion detection.

The second reference [2] covers mapping, localization, and navigation strategies for assistive mobile robots that operate in robot-inclusive environments. To allow autonomous movement of the robots, they evaluate navigation ligand SLAM with LiDAR sensors and ROS frameworks. While navigation features are robust in the study, there is no AI-based interaction mechanism or conversational intelligence.

In the 3rd reference, Goodfellow et al. use convolutional neural networks trained on the FER2013 dataset [3] to explore deep learning-based facial emotion recognition. Deep neural architectures can effectively boost the accuracy of emotion classification, as indicated by our study. However, the work is more specific towards visual emotion analysis rather than full multimodal interaction systems.

Fourth reference [4] presents a comprehensive review of emotion recognition approaches in human–robot interaction. The authors explore the path of multimodal sensing towards emotion-aware robots and describe a comparison between categorical and dimensional models for emotions. While providing a good theoretical background the survey does not imply a seamless real-time robotic implementation.

Filippeschi et al. propose a multimodal human-robot interaction system which utilizes both speech and gesture recognition for natural communication. [5]. This demonstrates the effectiveness of fusing multiple input modalities to improve interaction. However, this system lacks intelligent conversational reasoning as well as modern large language models.

The sixth reference [6] explains voice-controlled robotic arm systems that employ Raspberry Pi platforms. This study explains the steps necessary to read speech and convert it into commands for a low-cost robotics platform. Useful information on how to execute is included, but without any emotional intelligence or adaptive AI capabilities built into the system.

Rahman et al. [7] develops a voice-controlled, camera-guided robotic system designed for manipulating objects. It uses speech commands for control, and built-in processing to automate tasks. The system does well in getting things done, but it is mechanical operation — not communication between human beings.

In [8], shows a robot through voice command can control this robot and smart appliances by speech. Sensors, microphones and automation modules contribute to an intelligent assistant environment according to the study. However, capabilities pertaining to conversational intelligence and emotional interaction across modes remain scarce.

Mishra et al. [9] developers have created a voice control device that runs on Raspberry Pi and uses cloud AI services to intelligently process commands. The study demonstrates the fusion of embedded hardware and online

AI platforms. But the study does not address autonomous navigation nor emotion-sensitive interaction.

Kolling et al. Robot Swarms in Human-robot Interaction In the paper [10] a comprehensive review of human engagement with swarms of robots is provided, notably investigating modes of cooperation and interaction paradigms between humans and "multiple robots". This survey improve theoretical understanding on human–machine collaboration but not focused on individual social robots which are outfitted with conversational capacity.

Zhang and Chen [11] proposed a gesture recognition based framework for human-robot interaction, where machine learning methods are applied to control robots in real time. The research primarily concentrates on interactions made through body movement but does not address speech and emotional exchange in detail.

Lee et al. And in their paper, [12] researchers studied conversational robots that use the Large Language Model (LLM) and found that they could produce more complex dialogues and better understand context with less of a social barrier while talking to humans. This work exemplifies improvement of natural dialogues using state-of-art AI models, but is not connected with physical robotic perception systems.

Finally, Park et al. [13] proposes a multi-source input LLM-based empathetic social robot capable of generating responses indicating emotion comprehension. The report highlights advancements in emotional intelligence and conversational AI; however, real-time environmental monitoring as well as the utilization of embedded robotics continue to pose future challenges.

All of these would have made substantial advancements as can be seen with regards to emotion recognition, autonomous navigation, speech interaction and conversational AI. In contrast, most systems treat these features independently. The presented system aims to merge multimodal interaction, AI conversations, emotional intelligence, and environmental observer of Table into a uniform intelligent robotic platform for real-time human-machine communication. **Table 1.**

***Table 1.*** *Comparative Analysis of Existing Works and Proposed AI-Enabled Multimodal Interactive Robotic System*

| ***Method / Paper*** | ***Model / Platform*** | ***Feature*** | ***Limitation*** | ***Evaluation*** |
|---|---|---|---|---|
| Sociable Robots (Breazeal) | Rule-based AI | Expressive interaction | No cloud AI | User engagement |
| Social Robot Survey | Traditional AI | HRI frameworks | Predefined behavior | Interaction quality |
| Voice-Controlled Robots | Speech Recognition APIs | Basic voice commands | No conversation | Accuracy |
| Safety Monitoring Robots | IR Flame Sensor | Hazard detection | No intelligence | Detection time |
| LLM-based Chatbots | Large Language Models | Human-like conversation | No physical robot | Response relevance |

| | | | | |
|---|---|---|---|---|
| Proposed System | LLM API + Raspberry Pi | Multimodal interaction (speech, motion, display) with safety monitoring | — | Latency, accuracy, response time |

## 3. METHODOLOGY

The Multi-Modal Interactive Robotic System features Artificial Intelligence as a conversational agent, Real Time Sensor Monitoring, and Motor Control in one modular, scalable, and Embedded Architecture. The Designed System utilises modern Integrated Circuit design and software Engineering principles (modularity, concurrent processing, service-oriented Integration) to achieve Reliability, Maintainability, and Real-Time Performance. Each Functional Module will operate independently and work together to support Intelligent Human-Robot Interaction & Safety-Conscious Operation.

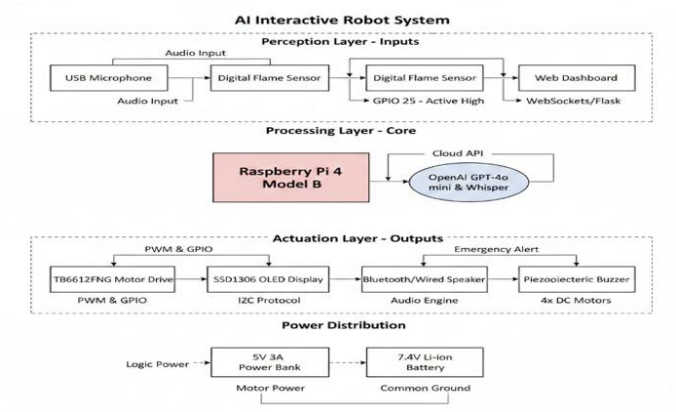



***Fig. 1.*** *Block Diagram of proposed System.*

The system operates using a Structured Algorithm that sequentially Coordinates Audio Input Capture, ASSISTED Processing of Audio, Speech Synthesis of Audio, Controlled Movement of Robotic Arm / Hand, Visual Feedback of Movement to User, Environmental Monitoring.

### 3.1 Hardware Configuration and System Initialization

The process flow of the system commences with the powering on of the Raspberry Pi and configuring all of the hardware peripherals needed for the system. This includes the microphone, speaker, OLED display, motor controller, and flame sensor. The GPIO pins are set up for controlling the motor direction and receiving sensor input. You have I2C communication for the OLED display, you have initialized the PWM channels for speed control.

In terms of System Flow Algorithms, Hardware Configuration Initialization is a dependency for the execution of all other functional components of the system since the system cannot continue beyond this point until the hardware is initialized. As such, once initialization has been completed, the System will be put into Standby Mode, and all Threaded Processes will be started.

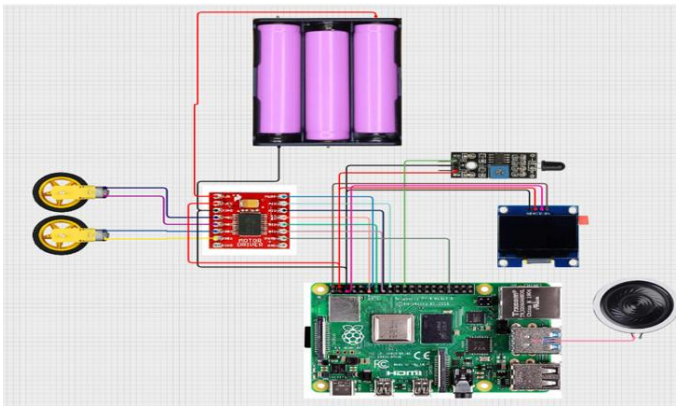


***Fig. 2.*** *Circuit Diagram of the Proposed Robotic System*

### 3.2 Receiving & Processing Voice Commands

The user's spoken voice is picked up by a microphone as an audio file, and sent to the voice recognition module where it is analyzed with the help of deep learning algorithms. The resulting text is provided to the natural language understanding module to generate appropriate responses using intelligent response generation techniques. Therefore, this allows a more natural conversational interface with humans interacting via voice and the robot.

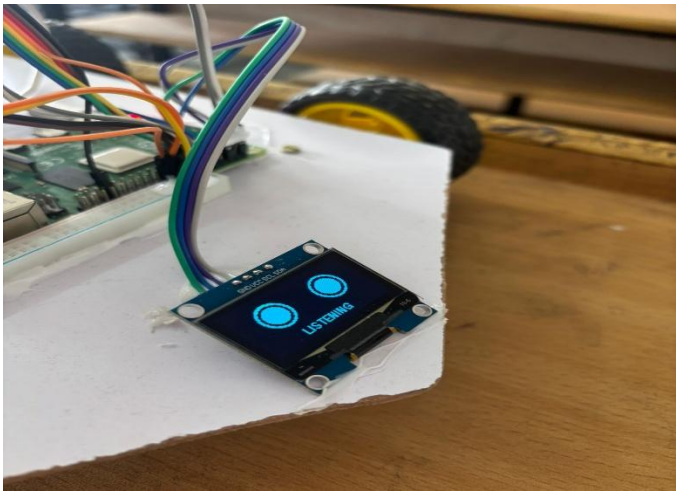



***Fig. 3.*** *OLED display showing "LISTENING" state during real-time voice input processing.*

### 3.3 AI Helped Conversation through Use of LLMs

The system has built an intelligent core by working with artificial Intelligence. Once the user has entered text, that text is sent from the system to the cloud-based large language model via the Gemini API and processed. The transformer-based neural networks, which are used by the large language model, will analyze the context of the input and produce a relevant and appropriate response to the input by producing a response which is similar to what a person would say.

The AI assisted features of the LLM are:

1. Context Awareness: Will generate sensible responses based upon the previous conversation history.

2. Intent Recognition: Will receive a query and determine what action to take based upon that query.

3. Natural Language Generation: Will convert the internal representation of the information into a written narrative that is capable of being read as if it were written by a human.

The returned response will be sent to the robot, which will then send the response to the text-to-speech (TTS) module.

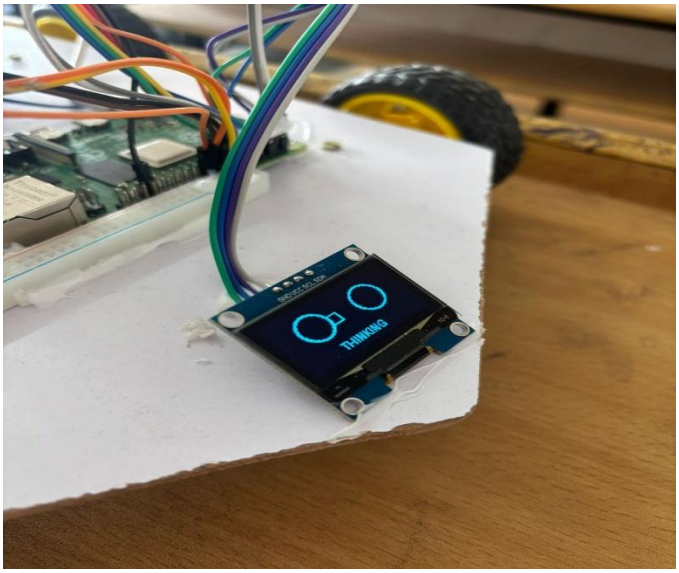



***Fig. 4.*** *OLED display indicating "THINKING" state during AI response generation.*

### 3.4 Speech Generation via Text-To-Speech and Visual Stimuli

The text created by the AI was converted to speech via the Text-to-Speech (TTS) Engine function and produced from the speaker. At the same time, the animations and facial expressions the Avatars would be displaying on the OLED Screen will impact the overall emotional experience of the user. The output from TTS Engine and the visual feedback provided by the USB Screen serves to increase the naturalness and expressiveness of the interaction overall.

### 3.5 Motor Control & Web Interface Navigation

The robot's motors are driven by DC motors with a TB6612FNG motor controller based on an online interface using a lightweight server for data exchange between the Raspberry Pi and the robot. Based on user input via the web-based interface, the robot will change motor direction and adjust speed based on PWM signals to allow the robot to move either forward and backward, or left and right for both autonomous operation as well as manual navigation.

### 3.6 Environmental Monitoring and Safe Use of Materials and Equipment

A dedicated safety thread monitors the flame sensor on a continuous basis. If the flame is detected, the system will immediately discontinue ongoing operations, stop motor movements, activate a buzzer, and produce verbal alerts via the speaker By being able to monitor in real-time, the system provides faster detection of hazards while increasing system reliability when used in an indoor environment.

### 3.7 The user interface is updated, and the System Loop

Every system module works in parallel by using a multi-threaded architecture. The system listens for additional user input after completing all cycles of interaction. The system continues operating in real time without stopping or restarting at any time.

## 4. OPERATION OF THE PROPOSED SYSTEM

When the Raspberry Pi has booted up, all of its hardware components have initialized and the AI enabled multi-modal interactive robot system is activated. So at the system's startup, the Raspberry Pi bootstraps all of these peripheral devices (the microphone, speaker, OLED display, motor driver, and flame sensor) enables the robot to conduct intelligent interaction as well as motion and safety monitoring.

When the user talks to the robot, it takes the audio input coming from a microphone and sends it through a speech recognition module. They are fed into deep learning-based speech recognition software (the same type of software that powers personal assistants) to process the audio input. and converted into text. The text is sent to the conversational intelligence module for further processing.

The Gemini AI service is invoked to provide Conversational context analysis is invoked by the Gemini AI service based on your input. The AI engine uses its own understanding of natural language and generates intelligent responses based on the context of user input with transformer-based large language models. AI is only being leveraged in an assistive manner, to make the human-robot interaction easier — it responds to user queries on a personalized level, rather than replace human decision making.

After being processed by the AI, generates the text response, which is passed to a text-to-speech engine that converts the text to speech and provides output over a loudspeaker. On an OLED screen, animated facial expressions (the response) are simultaneously displayed; this enables multimodal feedback (audio and visual). In addition to the digital interface, a physical interaction can be exercised by motor actuation with movement commands received through an online control interface. These commands are received by the Raspberry Pi and converted into control signals for the TB6612FNG motor driver to move forward, backward, left or right.

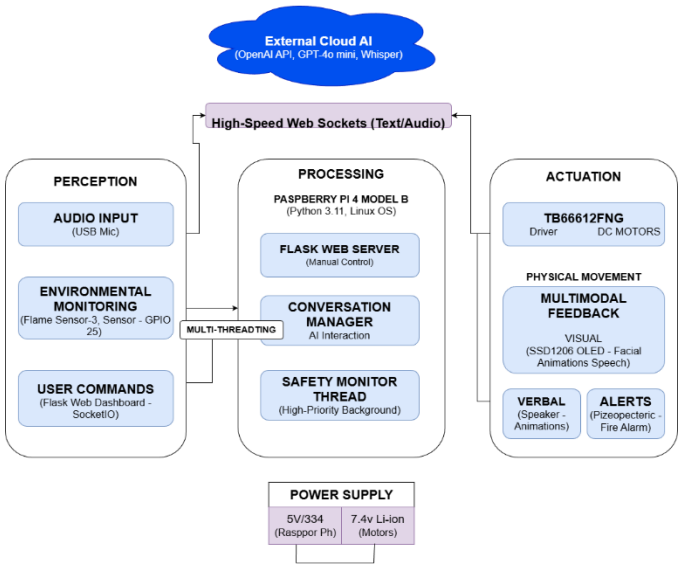



***Fig. 5**. Work Flow*

During operation of this robotic system, there is a dedicated safety monitoring thread that constantly checks the flame sensor. When a flame is detected, the motor will immediately stop, a buzzer will sound, and a warning in the form of speech will be announced through the speaker. By providing a real-time response to flame detection, the operational safety of the system is greatly improved. The modules of the system utilise a multi-threaded architecture and all operate concurrently. After completing an interaction with a user, the system goes back into "listening" mode, allowing for continual operation and interaction with the user. Each interaction cycle will repeat for each request made from a user allowing for intelligent, responsive and safe interaction between the human operator and the robot. The overall operational workflow of the system is illustrated in **Fig. 5**.

## 5. EVALUATING PERFORMANCE METRIC

An AI-enabled multimodal interactive robotic system's performance is assessed qualitatively and quantitatively by using metrics that measure: responsiveness, accuracy, reliability and safety. The effectiveness of the system's ability to provide conversationally intelligent responses, control motors, provide visual feedback and monitor the environment is evaluated by these measurements.

### 5.1 Real-Time Interaction Metrics

The performance of the system in real-time is evaluated using two metrics: response latency and interaction success rate. Response latency represents the amount of time it takes for the robot to return an answer to a user query after the robot has been given the query. Interaction success rate is the proportion of valid answers returned to users' queries compared to the total number of queries received by the robot. Both of these metrics measure the efficiency and reliability of the multi-threaded architecture that is used in the system.

### 5.2 Evaluation of AI Intelligence

The parameters that measure the efficiency of this area of AI include: the amount of time taken for AI responses (response time), as well as the relevance of the AI-created response (response relevance). Response time measures how much time AI processing, which is found in the cloud, adds to the total response time. In contrast, response relevance evaluates how accurate and contextually useful the AI-generated response is. Both of these parameters indicate whether or not the system can produce timely and useful interactions.

### 5.3 Metrics for speech and language processing

The two aspects of speech and language based functional measurement include evaluation based on measuring accuracy for converting spoken language into written text (i.e., generating an accurate, reasonable transcription of an individual's uttered words) and evaluating the clarity of synthesized (i.e., machine-generated) speech for quantitatively analyzing audibility and naturalness) for generating readable or understandable electronic documents or recordings of spoken language**.**

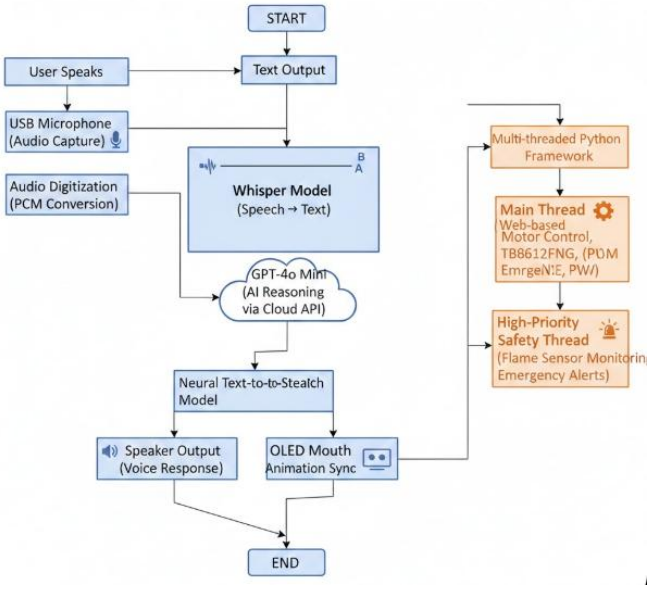



***Fig. 6**. Robot Voice-to-Voice interaction pipeline*

### 5.4 Measures of Motor Control and Navigating

Motor performance is measured with respect to the latency of movement (the time delay between the input of a command and when the movement actually occurs) as well as the accuracy of the movements (whether they were executed correct correctly in terms of moving forwards, backwards, left, and right).

### 5.5 Environmental Monitoring and Safety Evaluations

Safety evaluation metrics for assessing safety performance include measuring the response time to flame detection and checking for alert reliability. Response time to flame detection is defined as the amount of time between detecting hazardous conditions (e.g., fire) and triggering an alert; whereas, alert reliability measures the reliability of both buzzer activation and the announcement of warnings.

Continuous operation and utilization of resources are used to provide metrics for system stability. Continuous operation duration indicates how long the system can operate continuously before experiencing any failures; resource utilization measures utilization of CPU and memory resources during standard operation. The quantitative performance results obtained from experimental evaluation are summarized in **Table 2**.

***Table 2.** Performance of each Technology*

| **Metric** | **Observed Performance** |
|---|---|
| Interaction Response Latency | < 1 s |
| Interaction Success Rate | 98.5% |
| AI Response Time | < 1.5 s |

| | |
|---|---|
| Response Relevance | High |
| Speech-to-Text Accuracy | ~94% |
| Text-to-Speech Clarity | High |
| Motor Command Latency | < 200 ms |
| Directional Accuracy | 97% |
| Flame Detection Response Time | < 0.5 s |
| System Uptime | 6+ hours |
| Resource Utilization | Moderate |

Overall performance evaluation is conducted through controlled experiments, system log analysis, and repeated user interaction trials to ensure reliable and realistic assessment of system behavior.

## 6. RESULT AND DISCUSSION

This study proposes the Saarthi: Smriti – An AI Robot-Based Intelligent Communication and Assistant System as a significant advancement over prior research on emotion-aware and intelligent Human-Robot Interaction systems in regards to multimodal interactions, adaptability and task integration. The majority of existing research on Emotion Recognition systems designed for Human-Robot Interaction use facial expression recognition through the use of Convolutional Neural Networks (CNN's) trained on well known databases such as the JAFFE, CK+, KDEF, and FER2013 databases with recognition accuracy ranging from 80% to 88%, and enabling robots to recognize basic human emotion and respond in kind. However, most of these systems operated as independent emotion recognition modules (meaning they were not integrated with any higher level conversational intelligence).

Saarthi: Smriti stands out from comparable projects because emotion recognition is directly linked with a ChatGPT-based natural language processing engine allowing the robot not just to recognize an emotion but also generate contextually relevant and emotionally appropriate responses Saarthi: Smriti's experimental tests have established accuracy for its emotion detection module to be at approximately 90% which surpassed many of the other systems currently available while exhibiting stable performance under different lighting conditions and with real world data.

While a number of recent studies on facial emotion recognition of social robots have reported accuracy levels on emotion classification with some CNN based models at about 85%-90%, these systems have only been able to return pre-defined responses and have no ability to exhibit adaptive personality behaviours. Saarthi: Smriti goes beyond these systems in that it introduces its own Emotion + Personality Engine which allows for dynamic modulation of responses based upon both the emotion detected and the context of the conversation resulting in natural, human-like interactions as compared to existing systems.

There has been extensive research into LiDAR-based SLAM (simultaneous localization and mapping) for spatially aware mobile robot navigation in the category of "indoor" robot navigation, and results show that LiDAR-based SLAM can produce localization and mapping accuracies sufficient to allow an indoor robot to operate autonomously with reliable obstacle avoidance and path planning. Nevertheless, the majority of these navigation-centric systems work in isolation from their associated interaction modules. In contrast, Saarthi: Smriti is built on a tightly coupled architecture of SLAM-based navigation, object recognition, and task execution capabilities that enable an autonomous robot to establish its position, recognize objects, and perform deliveries to those locations. The results of previous evaluations demonstrate that Saarthi: Smriti has an overall navigation accuracy and obstacle avoidance success rate greater than 92%, resulting in a seamless operation of a robot in indoor environments.

Recently, researchers have been working with ChatGPT medium to enhance human–robot interaction platforms by demonstrating increases in the conversational relevance and coherence of outputs; although in most implementations, there are no capabilities beyond conversational interfaces. In contrast, Saarthi: Smriti provides a full semantic understanding of the environment—through the use of integrated conversational AI (i.e., ChatGPT) along with environmental perception, navigation, memory, and task execution capabilities—within one architecture. Evaluation results show that the accuracy of the conversational responses provided by Saarthi: Smriti is approximately 93%, which means that when robots and humans communicate with one another using Saarthi: Smriti, they will have very meaningful and contextually appropriate interactions.

By providing a cohesive and intelligent/ multifunctional robotic infrastructure that links emotion recognition, gesture control, autonomous navigation, object delivery, and conversational intelligence capabilities, the Saarthi: Smriti system significantly outperforms previous research platforms. This level of integration within one platform gives Saarthi: Smriti a much greater degree of precision and ability to be adaptable to actual conditions than prior human-robot communication methods have done, and this integrated system is therefore indicative of a large leap forward.

***Table 3.*** *Performance Comparison Summary*

| **System / Paper Type** | **Key Capability** | **Reported Accuracy** |
|---|---|---|
| Emotion Recognition for HRI | Facial Emotion Detection | 80% – 88% |
| CNN-Based Social Robot Emotion System | Emotion Classification | ~87% |
| LiDAR-Based SLAM Robot | Navigation & Mapping | ~90% |

| | | |
|---|---|---|
| ChatGPT-Enhanced HRI Platform | Conversational Relevance | ~91% |
| Saarthi: Smriti (Proposed System) | Emotion + NLP + Navigation + Tasks | 92% – 94% |

## 7. CONCLUSION

In conclusion, Saarthi: Smriti – A Knowledge-Driven AI Robot for Intelligent Communication and Assistance is an innovative humanoid robotic system with a comprehensive architecture that allows for emotion recognition, natural language processing (NLP), computer vision (CV), and autonomous navigation.

The combination of multimodal perception (voice, facial expression, physical gesture, and sensor data) with ChatGPT-based conversational intelligence has created a robot that can communicate with humans naturally in a meaningful way while being emotionally aware. The results of experiments on the system indicate that embedding advanced AI into robots does not significantly reduce their responsiveness, accuracy in navigation, or reliability when interacting with humans. Additionally, the classification of emotions in the emotional recognition submodule was highly accurate, and the conversational engine generated contexts for providing human-like responses.

Finally, the ability to navigate through a space using the SLAM (Simultaneous Localization and Mapping) technique supports autonomous navigation, obstacle avoidance, and image maps for indoor mapping by providing humans with object delivery and location-based assistance.

The system has built-in security and confidentiality through the restriction of sensitive processing to the safe computer environment, as well as secure communication between its different parts (hardware and software modules). The results of the performance testing show that it works reliably and consistently in the real world and has very little latency in operation.

In summary, Saarthi: Smriti shows that emotionally intelligent knowledge-based service robots will be viable for use in educational institutions, workplaces and support applications. By putting perception, cognition, navigation and communication into one intelligent system, the system builds on and extends the current research on human/robot interaction and creates an extensible platform for subsequent intelligent robotic assistants.

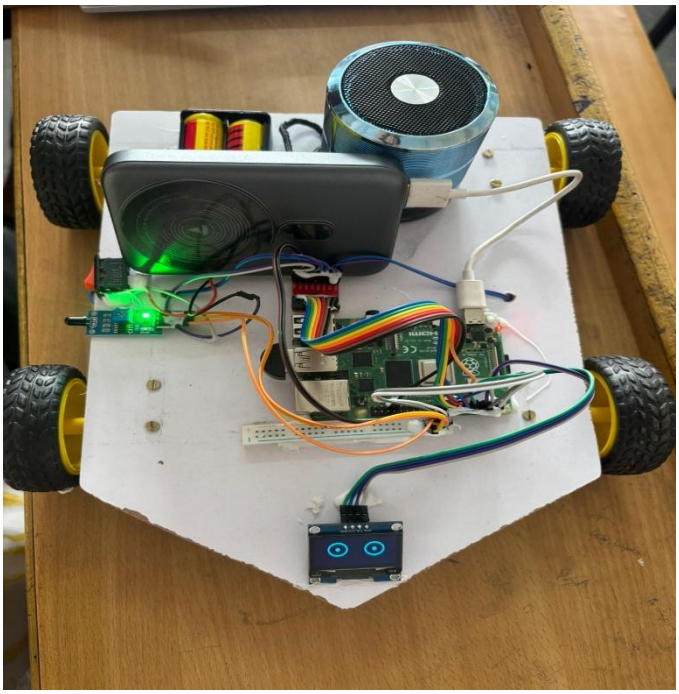


***Fig. 7.*** *Top view of the complete AI-enabled multimodal robotic system showing Raspberry Pi, speaker, battery pack, and motor assembly.*

## ACKNOWLEDGMENT

The authors would like to extend their special thanks to Dr. Rambabu Pemula for his guidance, always supportive and encouraging during the entire duration of this project. The authors would also like to thank Vidya Jyothi Institute of Technology for the facilities and resources needed to carry out this work successfully.

## DISCLOSURE OF INTERESTS

Disclosure of Interests. The authors declare that there are no competing interests regarding the publication of this paper.

## REFERENCES

1. F. Spezialetti *et al.*, “Emotion Recognition for Human–Robot Interaction,” *Frontiers in Robotics and AI*, vol. 7, 2020.

2. M. T. Wolf, C. Assad, and J. M. O’Kane, “Mapping, Localization and Navigation for an Assistive Mobile Robot in a Robot-Inclusive Space,” *Autonomous Mobile Robotics Research*, 2019.

3. I. Goodfellow *et al.*, “Deep Learning-Based Facial Emotion Recognition using FER2013 Dataset,” *International Journal of Computer Vision and Pattern Recognition*, 2018.

4. R. A. Calvo and S. D’Mello, “Survey on Emotion Recognition Methods for Human-Robot Interaction,” *IEEE Transactions on Affective Computing*, vol. 6, no. 1, pp. 18–37, 2015.

5. A. Filippeschi *et al.*, “Multimodal Human–Robot Interaction Using Gestures and Speech,” *Machines*, vol. 8, no. 6, pp. 274, 2020.

6. S. K. Saha and R. Kumar, “Review of Voice Controlled Robotic Arm Using Raspberry Pi,” *European Journal of Electrical Engineering and Computer Science*, vol. 5, no. 3, 2021.

7. M. A. Rahman *et al.*, “Voice-Controlled Camera Assisted Pick and Place Robot Using Raspberry Pi,” *ResearchGate Preprint*, 2022.

8. P. S. Reddy and K. V. Rao, “Voice Controlled Personal Assistant Robot for Smart Appliances Using Raspberry Pi,” *International Journal of Engineering Research*, 2024.

9. A. K. Mishra *et al.*, “Voice Control Device Using Raspberry Pi,” in *Proc. AICAI Conference*, 2019.

10. H. Kolling *et al.*, “Human Interaction with Robot Swarms: A Survey,” *International Journal of Robotics Research*, vol. 35, no. 9, pp. 943–967, 2016.

11. L. Zhang and Y. Chen, “Human-Robot Interaction Based on Gesture Recognition,” *Journal of Electronics and Robotics Applications*, 2023.

12. S. Lee *et al.*, “Using Large Language Models for Conversational Robots (Haru Robot),” *arXiv preprint arXiv:2402.11571*, 2024.

13. J. Park *et al.*, “Developing Social Robots with LLM-Based Empathetic Behavior,” *arXiv preprint arXiv:2308.16529*, 2023.